Agentic FEA of the Zhangjinggao Construction Catwalk:

Three hashed decks, the 41fb3222 connectivity table,
and a factorable but unconverged first increment

Run `cursor/catwalk-main-deck-bound-4c2c`
Repository `Hiram-test/model`, pull request #19

2026-08-25

Abstract

This paper records a complete, auditable agentic finite-element modelling run for the Zhangjinggao construction catwalk. A 77 MB millimetre centre-line STEP is mapped to $x = \text{chainage} - K16+876.000$. Floor-rope and portal-rope anchors are disjoint `*NSET`/`*BOUNDARY` families. Twenty-one cross-passages and 142 portal frames (71 stations per deck; Chinese classifier U+6980, not U+698C) reconcile with zero insertions.

Frozen deck `82548e6a3bd2612b6b39a08c313402b32a1961af6eba018158267906276ab6da` keeps its illegal ELSET+uniaxial `TYPE=STRESS` card and is not rewritten. Site `41fb32225489b0c6f993d3a077ce929` is the IC-pass / first-increment-singular deck and is not rewritten. An independent reread of that site recovers 22 096 used-node components and 19 371 two-node fragments = 17 756 T3D2 + 1 615 B31. The integer 21 426 counts **components that contain no constrained node**, not unconstrained nodes (50 676). The three-card boundary union is 1 220.

The new main deck `c635dad78661e6495bdf829b0c97f8610b5fedb8603307e806860032a70cda84` attaches drawing-anchor stubs at $x = -23.895/-44.909/4225.7$, consumes T3D2 yarn gaps and B31 pairs, pins leftover unconstrained *components*, and writes 421 432 eight-field second Piola–Kirchhoff rows. CalculiX 2.21 reads it, assembles 1 401 126 equations, factors the tangent (not singular), and fails to converge the first increment (exit 201, 63.65 s). The `.frd` file is an 80-byte title shell with no DISP. No spectrum is claimed.

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1 Introduction

Related work. Automated CAD-to-FEM pipelines usually inherit one coordinate system and one support family. A multi-span suspension-bridge construction catwalk does not: two walkways, two rope families with different physical anchor stations, discrete portal frames, and discrete cross-passages, all written in highway chainage. Theory archive `catwalk-theory/thirteen-mode-models` v1.2 freezes drawing topology and the fourteen-mode isolation protocol; it is not a CalculiX deck. Pull request #18 shipped an empty `artifacts/`. Pull request #19 first deposited `82548e6a`, then `41fb3222`. This run does not rewrite those two sites. It emits a third hash.

Method. The run follows `catwalk-fem/SKILL.md`. Geometry is STEP-only. Materials, sag, and loads come from drawings and the check report. The coordinate gate is identity unless saddle-height evidence requires a shift, never $X - x_{\min}$. Topology is reconciled against frozen drawing lists: 21 passages and $71 \times 2 = 142$ portal frames. Anchors are two families. `82548e6a` and `41fb3222` are frozen. The new main deck is written to `zjg_catwalk_main.inp`. CalculiX runs only on a copy. A successful read, a factorable tangent, or a `.sta` increment row is not a solved statics claim. Solved means exit 0, DISP in `.frd`, and a converged increment row.

Experiment. The centre-line STEP (139 991 trimmed curves, `SI_UNIT(.MILLI.,.METRE.)`) is converted to metres. After identity mapping, STEP nodes satisfy $X \in [0, 4270.609]\,\mathrm{m}$ and $Z_{\max} = 350.312\,\mathrm{m}$. Three SHA-256 values were rechecked on this host. CalculiX 2.21 on a copy of `c635dad7` assembled 1 401 126 equations and failed to converge the first increment.

Outlook. The new hash is emitted. The first increment is no longer immediately singular because of 22 096 components. Static convergence still requires work on the remaining 1 073 components, 477 pair fragments, four unconstrained components, and the `umpc_mean_rot` failure—not a rewrite of the two frozen sites. TARGET-FREQ stays isolated until a solved spectrum exists.

2 Related work and isolation protocol

Related work. Agentic FEA here means a scripted chain with frozen constants. CalculiX 2.21 is used because the keyword deck is inspectable. Abaqus-like ELSET-wide uniaxial prestress is not the CalculiX 2.21 contract.

Method. Isolation rules from theory v1.2 §1 and the skill file:

1. STEP supplies coordinates only.
2. Properties come from DRW-A/B, CALC-INPUT, STD, or a named ASSUMP.

3. `TARGET-FREQ` lives in `catwalk-fem/isolated/` and is not imported by `write_inp` or the solver deck.
4. Formed sag is $h = 227.300\,\mathrm{m}$. The 255.56 m main-cable control sag is forbidden in the deck (verified absent).
5. The two walkways keep independent degrees of freedom.
6. Floor-rope anchors and portal-rope anchors are never unioned.
7. After SHA-256 `82548e6a` and `41fb3222` are frozen, those `.inp` files are not rewritten to "make the solver happy".

Experiment. The isolated file lists fourteen frequencies starting at 0.0296 Hz. Full-text search of all three decks finds neither 0.0296 nor 255.56. `write_inp.py` does not import `isolated/TARGET-FREQ.json`.

Outlook. Isolation is a scientific claim. Later comparison tables may load TARGET-FREQ only after deck, solve, and mode-shape classification are frozen.

3 Charter, evidence grades, and frozen constants

Related work. Evidence grades follow theory v1.2: DRW-A/B, CALC-INPUT, STD, TEST, ASSUMP. Portal and passage stations are DRW-B dimension chains. Saddle coordinates are CALC-INPUT (check-report tables 1-5 / 1-9).

Method. The charter freezes project `zjg-catwalk`, SI internal units, and four load cases LC-DEAD-PRESTRESS, LC-PERSONNEL-UNIFORM, LC-WIND-Y, LC-FREQ. The origin is the nominal four-span north end, not a physical anchor:

$$x = \text{chainage} - K16 + 876.000, \qquad y \text{ upstream} \to \text{downstream}, \qquad z \text{ up}.$$

Physical anchors stay in four named stations:

$$\begin{aligned} x_{\text{floor,N}} &= -23.895 \text{ m } (K16 + 852.105), & x_{\text{floor,S}} &= 4210.368 \text{ m } (K21 + 086.368), \\ x_{\text{portal,N}} &= -44.909 \text{ m } (K16 + 831.091), & x_{\text{portal,S}} &= 4225.700 \text{ m } (K21 + 101.700). \end{aligned}$$

Nominal four-span breaks $\{0, 660, 2960, 3677, 4180\}$ and physical saddles $\{666.679, 2953.321\}$ are registered together.

Experiment. All three headings state the convention in plain text. `82548e6a`/`41fb3222` have STEP $x_{\min} = 0$. After drawing stubs the new main has $x_{\min} = -44.909$; STEP-derived nodes still start at 0. No $X - x_{\min}$ shift is applied.

Outlook. If a later STEP covers the north physical anchors, the four stations remain four sets. Stubs are attachments, not a mixed floor/portal set.

4 Ingestion and classification (N02–N07)

Related work. A centre-line STEP of 139 991 trimmed curves is not a member-role model.

Method. `parse_step` stream-parses `CARTESIAN_POINT / TRIMMED_CURVE`, checks the STEP SHA-256, and scales millimetres to metres. `classify_segments` labels floor ropes by the drawing $y = \pm(21.45 \pm (0.85 + 0.26k))$ lattice, long high members as portal/handrail ropes, short transverse members as `portal_or_beam`, and long inter-deck members as `cross_passage`.

Experiment. Raw counts: `floor_rope` 43 200, `portal_or_beam` 4 719, `portal_rope` 356, `cross_passage` 0, `short_other` 91 618. No single segment has $\Delta y \geq 15$ m, because each 49.655 m passage is tessellated into ≈ 1.7 m pieces. Detection therefore uses Y-span. Unclassified `short_other` is dropped from the coarsened deck.

Outlook. Portal-rope classification remains incomplete after coarsening. That bound is reported in §10 and is not hidden by the 142 frame count.

5 Coordinate and topology gates (N07/N09)

Related work. An empty `artifacts/` cannot be a passed coordinate gate. Topology counts must be recovered from the mesh against drawing lists.

Method. `infer_x_transform` scores identity, minus-$x_{\min}$, and minus raw chainage. Alignment is confirmed by local Z_{p90} within 12 m of $x = 666.679$ and $x = 2953.321$. Passages: 21 drawing stations. Portal frames: 71 stations $\times$ 2 decks = 142. The Chinese unit is U+6980; U+698C is not used.

Table 1: Coordinate and topology gates.

Check	Result	Evidence
Units	PASS	mm STEP → m nodes; $x_{\max} = 4270.609 < 20\,000$
Transform	PASS	identity, shift $= 0$
Saddle Z	PASS	north $Z_{p90} = 330.78$ m, south 324.85 m vs 340.6 m
Two decks	PASS	Y medians $+20.6$ / -20.6 m vs ± 21.45
Formed sag	PASS	geometry 214.18 m vs 227.30 m, $\lvert\Delta\rvert = 13.12$ m
No $x_{\min}$ shift	PASS	STEP $x_{\min} = 0$; drawing stubs may have $x < 0$
Passages	PASS	21/21, insertions 0
Portal frames	PASS	142/142, insertions 0

Experiment. Twenty-one passages close by Y-span at every DRW-B station (Table 2). Portal frames close station-by-station (Table 5).

Outlook. The 13.12 m sag residual is inside the 15 m gate. It is not a formed-line fit. Later runs must not treat histogram peaks as saddle evidence.

6 Materials, family anchors, and initial tension (N08–N10)

Related work. Theory v1.2 §4 forbids treating floor-rope anchors as portal-rope anchors. The two north stations differ by 21 m; the two south stations differ by 15.332 m.

Method. Wire rope $E = 1.20 \times 10^{11}$ Pa, floor/portal $A = 1400.42 \times 10^{-6}$ m^2, $\mu = 12.038$ kg/m. Steel $E = 2.06 \times 10^{11}$ Pa, $\rho = 7850$ kg/m^3. Horizontal force

$$H = \frac{wL^2}{8h}, \quad h = 227.300 \text{ m}, \quad L = 2286.642 \text{ m}$$

gives $H_{\text{floor}} = 7.954$ MN and $H_{\text{portal}} = 2.039$ MN, hence $\sigma_{\text{floor}} = 3.549611 \times 10^8$ Pa and $\sigma_{\text{portal}} = 2.426295 \times 10^8$ Pa. Floor and portal anchors use two `*BOUNDARY` cards. A third card pins saddles and nominal ends. The new main adds a fourth card for leftover unconstrained *components*; that set is not unioned into the anchors.

Table 2: Twenty-one cross-passages (DRW-B dimension chains).

No.	Label	x (m)	Chainage
1	P01	159	K17+035.000
2	P02	330	K17+206.000
3	P03	501	K17+377.000
4	P04	838	K17+714.000
5	P05	1009	K17+885.000
6	P06	1180	K18+056.000
7	P07	1351	K18+227.000
8	P08	1504	K18+380.000
9	P09	1657	K18+533.000
10	H1	1810	K18+686.000
11	H2	1963	K18+839.000
12	H3	2116	K18+992.000
13	H4	2269	K19+145.000
14	H5	2440	K19+316.000
15	H6	2611	K19+487.000
16	H7	2782	K19+658.000
17	H8	3138.5	K20+014.500
18	H9	3318.5	K20+194.500
19	H10	3498.5	K20+374.500
20	H11	3843	K20+719.000
21	H12	4014	K20+890.000

Experiment. On `41fb3222`, north floor and north portal remain STEP-end proxies ($x = 0$ and $x \approx 46.4$). On `c635dad7`, drawing stubs place FLOOR_N at -23.895, PORTAL_N at -44.909, and PORTAL_S at 4225.7. South floor remains a match ($\Delta = 0.38\,\mathrm{m}$). In both decks $N_{\mathrm{floor}} = 312$, $N_{\mathrm{portal}} = 16$, intersection empty.

Outlook. Stubs write drawing $x < 0$ into the new main only. They do not rewrite `41fb3222`.

7 Load cases (N11)

Related work. CalculiX does not inherit `*DLOAD`/`*CLOAD` across steps.

Method. The check-report per-deck package is split equally onto 16 explicit floor ropes so that $\sum_i w_{\mathrm{rope}} L_i = w_{\mathrm{deck}} L_{\mathrm{deck}}$.

Experiment. On the new main: extra dead 10.756 MN, personnel 103.006 MN, wind 6.131 MN. Floor-truss length after yarn stitch is 196 202 m. LC-FREQ requests 20 eigenvalues. The deck text contains neither 0.0296 nor 255.56.

Outlook. The long floor length is a classification/stitch bound, not an invented live load.

8 The 41fb3222 connectivity table (independent reread)

Related work. A previous write-up called `41fb3222` the "new main deck" and attributed singularity to 22 096 components. The component count holds. The integer 21 426 is easy to misread as an unconstrained-node count. The correction: 21 426 is the unconstrained-*component* count.

Method. `pipeline/singular_audit.py` reads `zjg_catwalk_ccx221.inp` only. Connectivity is the used-node element graph. Constraint nodes are the union of three `*BOUNDARY` cards: `N_FLOOR_ANCHOR` $\cup$ `N_PORTAL_ANCHOR` $\cup$ `N_SUPPORT_SADDLE_ENDS`. A component with empty intersection against that union is an unconstrained component. Two-node fragments are split by element type. The file is not rewritten.

Experiment. Host `sha256sum` equals `41fb3222`...`bbca924a`. `test_singular_defs` passes. `match_given.all` is true.

Table 3: Independent reread of site `41fb3222`. 21 426 is not an unconstrained-node count.

Quantity	Value	Meaning
Used-node components	22 096	connected components
Two-node fragments	19 371	size-2 components
of which T3D2	17 756	floor 17 616 + portal_rope 136 + handrail 4
of which B31	1 615	portal_or_beam 1 594 + passage 21
Unconstrained *components*	21 426	no constraint node in the component
Unconstrained *nodes*	50 676	$51\,896 - 1\,220$
Three-card BC union	1 220	$312 \cup 16 \cup 904$; floor $\cap$ portal $= 0$
Nodes / elements	51 896 / 30 317	

Identities: $19\,371 = 17\,756 + 1\,615$, $50\,676 = 51\,896 - 1\,220$, $21\,426 \neq 50\,676$.

CalculiX 2.21 on a copy of `41fb3222`: IC accepted, 879 076 equations, SPOOLES singular, exit 255, wall 10.35 s. `.frd` is 80 B with no DISP. `.sta`/`.cvg` have no increment rows. Not a solution.

Bounds kept on the frozen site: T3D2 wrote eight integration points (C3D8I expansion; IC accepted); north floor $x = 0$ vs -23.895; north portal $x \approx 43.9$–48.8 vs -44.909; south portal 4221.093 vs 4225.7; B31 and the remaining 51 T3D2 had no IC.

Outlook. The table is the input to the new-main repair. It is not a reason to withhold `41fb3222`. Do not pin 21 426 nodes.

9 New main deck `c635dad7` and CalculiX 2.21

Related work. CalculiX 2.21 §7.76 requires, before the first `*STEP` and without `USER`: element number, integration point, and six global second Piola–Kirchhoff components. T3D2 expands like B31 to C3D8I (manual §6.2.35), so eight integration points are written.

Method. `repair_topology.py` rereads the `41fb3222` mesh and does not write that path:

1. Drawing-anchor stubs: 28 members at $-23.895/-44.909/4225.7$.

2. Floor yarn stitch: same role, side, and Y (3 decimals), consecutive X, $\Delta x \leq 15\,\mathrm{m}$. 20 948 T3D2 members. This is the 17 756 T3D2 fragment path, not a 57–2000 m portal/handrail stitch.

3. B31 pair stitch only: same side, $|\Delta x| \leq 4\,\mathrm{m}$, $d \leq 3.5\,\mathrm{m}$. 485 members. This is the 1 615 B31 fragment path.

4. Couple floating beams/passages/portal ropes to the floor: 724+21+156.
5. Pin leftover unconstrained *components* as `N_ORPHAN_UNCONSTRAINED`: 1 012 components / 3 596 nodes.

`write_inp.py` writes eight-field PK2 for every T3D2 and every B31. Floor and portal ropes use $S = \sigma n \otimes n$; other roles use zero PK2. `emit_bound_main_deck.py` refuses to touch the two frozen hashes. `emit_new_main_deck.py` now refuses to overwrite `41fb3222`.

Experiment. Independent reread: 51 924 nodes, 52 679 elements, 421 432 legal IC rows, 0 ELSET+uniaxial rows, 1 073 components, 477 remaining pairs (110 T3D2 + 367 B31), 4 unconstrained components under all BC cards, three-card union 1 232, 29/29 gates PASS. Heading text states the `41fb3222` table, including "21426 = components with ZERO constraint nodes".

CalculiX 2.21 on `/tmp/ccx-c635dad7`:

```
This is Version 2.21
copy_sha256 = c635dad7...70cda84
parse_fail_ic = false
number of equations = 1401126
singular = false
exit 201
wall_s = 63.65
has_disp = false
n_sta_increment_rows = 1
solved = false
```

Table 4: Solver four-file record on copies. Not a solved statics claim.

Hash	Eq.	Sing.	Inc. rows	DISP	exit	Wall (s)
`82548e6a`	—	—	0	no	201	< 1
`41fb3222`	879 076	yes	0	no	255	10.35
`c635dad7`	1 401 126	no	1 (`1U`)	no	201	63.65

The new-hash `.sta` line is `1 1 1U 2` (unconverged). The `.cvg` file has four iteration rows with residual 10^7–10^8 % and 100 % displacement correction. Stdout ends in `no convergence` / `divergence` / `*ERROR in umpc_mean_rot` (node 78746). The `.frd` file is 80 B. After the run, both frozen hashes were unchanged.

Outlook. The lexical IC gate and "first increment no longer immediately singular" are closed. Convergence is the next bound. Do not change the §7.76 row format again, and do not rewrite the two frozen sites.

10 Discussion and conclusion

Related work. PR #18 shipped an empty directory. PR #19 deposited `82548e6a` then `41fb3222`. This run emits `c635dad7` from the corrected definition table and leaves traces under `catwalk-fem/eval/`.

Method. Visible limits are bounds. Hard gates (identity, 21/142, disjoint anchors, §7.76 lexicon, two frozen hashes, forbidden sources, the 21 426 definition) close. An unconverged first increment is a bound, not a reason to withhold the new deck.

Experiment. Bounds: `41fb3222` north anchors remain STEP-end proxies; the new main stubs north floor/portal and south portal; south floor still $\Delta = 0.38$ m; portal-rope classification is incomplete; geometric sag 214.18 vs 227.30 m; floor-line length is long; 1 073 components and 477 pair fragments remain; orphan pins are leftover unconstrained components, not a statically determinate weld; the new-hash tangent factors but the first increment does not converge; no DISP; no spectrum.

Outlook. From the published STEP, under $x = \text{chainage} - K16 + 876.000$, a readable CalculiX deck exists; floor and portal anchors are separate; 21 passages and 142 frames reconcile with zero insertions. Frozen `82548e6a` remains ELSET+uniaxial. `41fb3222` remains the IC-pass / first-increment-singular site; the independent table (22 096 / 19 371 = 17 756+1 615 / 21 426 components / 50 676 nodes / 1 220 union) matches the given arithmetic and is not rewritten. New `c635dad7` emits drawing stubs, yarn/pair consume, component pins, and 421 432 eight-field PK2 rows. CalculiX 2.21 reads it and assembles 1 401 126 equations, then fails to converge. No solved spectrum is claimed. 21 426 is not an unconstrained-node count.

Reproduction

```
python3 catwalk-fem/tests/test_coord_gate.py
python3 catwalk-fem/tests/test_write_inp.py
python3 catwalk-fem/tests/test_reconcile.py
python3 catwalk-fem/tests/test_audit_frozen_deck.py
python3 catwalk-fem/tests/test_new_main_deck.py
python3 catwalk-fem/tests/test_singular_defs.py
python3 catwalk-fem/tests/test_repair_topology.py
python3 catwalk-fem/pipeline/singular_audit.py \
    catwalk-fem/artifacts/zjg_catwalk_ccx221.inp
python3 catwalk-fem/pipeline/emit_bound_main_deck.py
sha256sum catwalk-fem/artifacts/zjg_catwalk_coarsened.inp
sha256sum catwalk-fem/artifacts/zjg_catwalk_ccx221.inp
sha256sum catwalk-fem/artifacts/zjg_catwalk_main.inp
pdflatex -interaction=nonstopmode zjg_catwalk_agentic_fea.tex
```

Do not add the 77 MB STEP to git. Do not feed `isolated/TARGET-FREQ.json` to the writer. Do not rewrite `zjg_catwalk_coarsened.inp` or `zjg_catwalk_ccx221.inp`. This run does not push.

Data availability

Artefacts live under `catwalk-fem/artifacts/` on branch `cursor/catwalk-main-deck-bound-4c2c`. Frozen `82548e6a...276ab6da`. Singular site `41fb3222...bbca924a`. New main `c635dad7...70cda84`. Evaluation traces: `catwalk-fem/eval/GROK_SELF_EVAL.md` and `eval/DEFINITION_TABLE_41fb3222.md`.

References

[1] G. Dhondt, *CalculiX CrunchiX USER'S MANUAL version 2.21*, 2023, §6.2.35 and §7.76.

[2] Zhangjinggao catwalk fourteen-mode independent theory archive, v1.2, `catwalk-theory/thirteen-mode-models`.

[3] `catwalk-fem/SKILL.md` and `catwalk-fem/PLAN.md`.

[4] Release `catwalk-attachment23-v2.0-s10-20260716`, `cw_S10_0716t050342_a4_centerline.step`, SHA-256 `d03d01e3...763344`.

A Station-by-station 142-frame ledger

Generated from `artifacts/portal_142_ledger.json`. Every station is present on both decks. Counting unit: U+6980, not U+698C.

Table 5: Station-by-station reconciliation of 142 portal frames (71 stations × 2 decks). Unit: frames (CJK U+6980), not U+698C.

No.	x (m)	Chainage	Span	Up	Dn	n_{up}	n_{dn}	OK
1	45	K16+921.000	N 660 m	Y	Y	5	5	Y
2	102	K16+978.000	N 660 m	Y	Y	8	8	Y
3	159	K17+035.000	N 660 m	Y	Y	8	8	Y
4	216	K17+092.000	N 660 m	Y	Y	31	35	Y
5	273	K17+149.000	N 660 m	Y	Y	8	8	Y
6	330	K17+206.000	N 660 m	Y	Y	8	8	Y
7	387	K17+263.000	N 660 m	Y	Y	32	35	Y
8	444	K17+320.000	N 660 m	Y	Y	9	9	Y
9	501	K17+377.000	N 660 m	Y	Y	8	8	Y
10	558	K17+434.000	N 660 m	Y	Y	34	38	Y
11	615	K17+491.000	N 660 m	Y	Y	9	9	Y
12	724	K17+600.000	Main 2300 m	Y	Y	8	8	Y
13	781	K17+657.000	Main 2300 m	Y	Y	8	8	Y
14	838	K17+714.000	Main 2300 m	Y	Y	8	8	Y
15	895	K17+771.000	Main 2300 m	Y	Y	24	29	Y
16	952	K17+828.000	Main 2300 m	Y	Y	8	8	Y
17	1009	K17+885.000	Main 2300 m	Y	Y	9	9	Y
18	1066	K17+942.000	Main 2300 m	Y	Y	25	30	Y
19	1123	K17+999.000	Main 2300 m	Y	Y	8	8	Y
20	1180	K18+056.000	Main 2300 m	Y	Y	8	8	Y
21	1237	K18+113.000	Main 2300 m	Y	Y	25	26	Y
22	1294	K18+170.000	Main 2300 m	Y	Y	8	8	Y
23	1351	K18+227.000	Main 2300 m	Y	Y	8	8	Y
24	1402	K18+278.000	Main 2300 m	Y	Y	45	50	Y
25	1453	K18+329.000	Main 2300 m	Y	Y	10	10	Y
26	1504	K18+380.000	Main 2300 m	Y	Y	10	10	Y
27	1555	K18+431.000	Main 2300 m	Y	Y	45	50	Y
28	1606	K18+482.000	Main 2300 m	Y	Y	10	10	Y
29	1657	K18+533.000	Main 2300 m	Y	Y	10	10	Y
30	1708	K18+584.000	Main 2300 m	Y	Y	44	49	Y
31	1759	K18+635.000	Main 2300 m	Y	Y	10	10	Y
32	1810	K18+686.000	Main 2300 m	Y	Y	10	10	Y
33	1861	K18+737.000	Main 2300 m	Y	Y	45	50	Y
34	1912	K18+788.000	Main 2300 m	Y	Y	10	10	Y
35	1963	K18+839.000	Main 2300 m	Y	Y	10	10	Y
36	2014	K18+890.000	Main 2300 m	Y	Y	45	50	Y
37	2065	K18+941.000	Main 2300 m	Y	Y	10	10	Y
38	2116	K18+992.000	Main 2300 m	Y	Y	10	10	Y
39	2167	K19+043.000	Main 2300 m	Y	Y	45	50	Y
40	2218	K19+094.000	Main 2300 m	Y	Y	10	10	Y
41	2269	K19+145.000	Main 2300 m	Y	Y	10	10	Y

Table 5: 142 portal frames (continued)

No.	x (m)	Chainage	Span	Up	Dn	n_{up}	n_{dn}	OK
42	2326	K19+202.000	Main 2300 m	Y	Y	29	32	Y
43	2383	K19+259.000	Main 2300 m	Y	Y	8	8	Y
44	2440	K19+316.000	Main 2300 m	Y	Y	8	8	Y
45	2497	K19+373.000	Main 2300 m	Y	Y	30	33	Y
46	2554	K19+430.000	Main 2300 m	Y	Y	9	9	Y
47	2611	K19+487.000	Main 2300 m	Y	Y	8	8	Y
48	2668	K19+544.000	Main 2300 m	Y	Y	32	37	Y
49	2725	K19+601.000	Main 2300 m	Y	Y	8	8	Y
50	2782	K19+658.000	Main 2300 m	Y	Y	9	9	Y
51	2839	K19+715.000	Main 2300 m	Y	Y	32	37	Y
52	2896	K19+772.000	Main 2300 m	Y	Y	8	8	Y
53	3018.5	K19+894.500	S 717 m	Y	Y	8	8	Y
54	3078.5	K19+954.500	S 717 m	Y	Y	9	9	Y
55	3138.5	K20+014.500	S 717 m	Y	Y	8	8	Y
56	3198.5	K20+074.500	S 717 m	Y	Y	8	8	Y
57	3258.5	K20+134.500	S 717 m	Y	Y	8	8	Y
58	3318.5	K20+194.500	S 717 m	Y	Y	8	8	Y
59	3378.5	K20+254.500	S 717 m	Y	Y	9	9	Y
60	3438.5	K20+314.500	S 717 m	Y	Y	8	8	Y
61	3498.5	K20+374.500	S 717 m	Y	Y	9	9	Y
62	3558.5	K20+434.500	S 717 m	Y	Y	8	8	Y
63	3618.5	K20+494.500	S 717 m	Y	Y	8	8	Y
64	3729	K20+605.000	S 503 m	Y	Y	8	8	Y
65	3786	K20+662.000	S 503 m	Y	Y	8	8	Y
66	3843	K20+719.000	S 503 m	Y	Y	8	8	Y
67	3900	K20+776.000	S 503 m	Y	Y	23	24	Y
68	3957	K20+833.000	S 503 m	Y	Y	8	8	Y
69	4014	K20+890.000	S 503 m	Y	Y	8	8	Y
70	4071	K20+947.000	S 503 m	Y	Y	22	24	Y
71	4128	K21+004.000	S 503 m	Y	Y	8	8	Y

B CalculiX 2.21 transcripts

Frozen hash (unchanged):

```
*ERROR reading *INITIAL CONDITIONS. Card image:
       E_FLOOR_ROPE,3.549611E+08
```

Singular site (unchanged):

```
number of equations
 879076
matrix found to be singular
```

New main:

```
number of equations
 1401126
no convergence
*ERROR in umpc_mean_rot
       node        78746
```

First legal stress row (both PK2 decks):

```
*INITIAL CONDITIONS, TYPE=STRESS
1, 1, 1.439957e+08, 0.000000e+00, 2.109655e+08, 0.000000e+00, 1.742932e+08,
    0.000000e+00
```